

4.To Split Input into Tokens, Identify Delimiters, Handle Whitespace, Create Token Structures, Validate Token Streams, Debug Parsing Output.

To Design Parser Logic, Generate Parse Trees, Validate Syntax, Detect Errors, Handle Empty Commands, Produce Execution Structures.

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#include <ctype.h>

#define MAX_INPUT 1024
#define MAX_TOKENS 100

int main()
{
    char input[MAX_INPUT];
    char tokens[MAX_TOKENS][MAX_INPUT];
    int tokenCount = 0;

    printf("==== Mini Shell Quote Parser =====\n");
    printf("Type 'exit' to quit.\n\n");

    while (1)
    {
        printf("MiniShell> ");

        if (fgets(input, sizeof(input), stdin) == NULL)
            break;

        input[strcspn(input, "\n")] = '\0';

        if (strcmp(input, "exit") == 0)
            break;

        if (strlen(input) == 0)
            continue;

        tokenCount = 0;

        int i = 0;

        while (input[i] != '\0')
        {
            while (isspace(input[i]))
                i++;

            if (input[i] == '\0')
                break;

            int j = 0;
```

```
prajwal@PRAJWAL:~/os$ cd OSSkill
prajwal@PRAJWAL:~/os/OSSkill$ nano skill4.c
prajwal@PRAJWAL:~/os/OSSkill$
prajwal@PRAJWAL:~/os/OSSkill$ gcc skill4.c -o skill4
prajwal@PRAJWAL:~/os/OSSkill$ ./skill4
===== Mini Shell Quote Parser =====
Type 'exit' to quit.

MiniShell> ls -l

Parsed Tokens:
Token 1 : ls
Token 2 : -l
-----

MiniShell> ls -a

Parsed Tokens:
Token 1 : ls
Token 2 : -a
-----

MiniShell> proc

Parsed Tokens:
Token 1 : proc
-----

MiniShell> |
```